

AI SYSTEM SPECIFICATION

NAME: Wareesha

ROLL NO: 2K23/CSE/153

SUBJECT: ARTIFICIAL INTELLIGENCE

ASSIGNMENT

1: AI SYSTEM SPECIFICATION

Assignment 1: AI System Specification

Chosen System: An AI Tutor for Personalized Student Learning Recommendation

Task 1: PEAS Specification

Performance Measures

Success for the AI tutor is not just about delivering content quickly, but also ensuring educational accuracy, student engagement, and knowledge retention. Some key metrics are:

- Recommendation accuracy (percentage of suggested topics/materials that match the student's actual learning gap)
- Average time taken by a student to master a recommended concept
- Student engagement rate (active platform usage time and assessment completion rate)
- Learning efficiency and improvement score (pre-test vs. post-test performance gains)
- System response delay or latency during real-time assessment processing

Environment

The AI tutor operates in a dynamic digital ecosystem accessed by diverse learners across different platforms.

The environment includes individual students, concurrent learners, external educational databases, learning management systems (LMS), and dynamic human inputs. Internet connection stability may change across different regions, and varying user devices or screen sizes can affect interface accessibility. The user context is semi-structured but changes rapidly based on real-time student interactions, mood, distraction levels, and progression speed.

Actuators

The system interacts by using the following mechanisms in the environment:

- Personalized dashboard interfaces to display customized learning paths
- Text-to-speech engine or chat interfaces to deliver guided natural language feedback
- Dynamic quiz generation modules to adaptively change question difficulty
- Notification systems to trigger reminders, motivation prompts, and alerts

Sensors

The AI tutor relies on multiple sensors and data streams to understand its learners' states and surrounding progress:

- Keystroke and click-stream loggers for analyzing platform navigation patterns
- Typing speed and timing tracking to detect hesitation or rapid guessing
- Text input fields and natural language parsers to interpret student queries
- Camera-based eye-tracking or emotion detection (where permitted) to assess attention levels
- Internal databases to track historical performance, past mastery metrics, and system health

Task 2: Environment Classification

- **Partially Observable:** The AI tutor cannot always see everything clearly, especially when the student's internal mental state, distraction level, or external assistance (e.g., cheating or offline help) is hidden.
- **Multi-agent:** There are multiple students and sometimes teachers or human tutors operating at the same time, all affecting the data trends and learning resource availability.
- **Stochastic:** The system is not fully predictable because user responses can have random errors, guessing factors exist, and human emotional behaviors are inherently unpredictable.
- **Sequential:** Each action taken by the AI tutor affects future decisions, such as which study material it recommends next or how it alters the difficulty of subsequent assessments based on current answers.
- **Dynamic:** The environment keeps changing continuously due to the student's evolving knowledge base, changing attention spans, and updated learning content uploaded to the system.
- **Continuous:** Metrics like time spent per question, engagement scores, and mastery probability happen over continuous values rather than discrete, fixed steps.

Task 3: Critical Analysis

1. Biggest Technical Challenge

The most difficult aspect is partial observation. In a live learning environment, the AI tutor never has complete information about its student's actual cognitive state or focus. Latent factors such as mental fatigue, offline distractions, or external help can remain hidden, while sensor inputs like timing data can occasionally be misleading. Because of this, the AI tutor must be able to make robust recommendations based on incomplete data, which increases the risk of recommending frustratingly hard or redundantly easy content, requiring advanced predictive modeling and student profiling techniques.

2. Utility Function (Learning Progress vs. Student Fatigue Trade-off)

A simple utility function can be defined as:

$$U = 0.6 \times \text{LearningGain} - 0.4 \times \text{StudentFatigue}$$

This means the AI tutor tries to balance maximize knowledge acquisition with minimizing student burnout and fatigue. Optimal recommendations improve learning efficiency, while excessive difficulty or volume increases fatigue, reducing overall performance.

If the weight of learning gain is doubled, the AI tutor would prioritize pushing intense or advanced material faster, even if it drastically increases the student's cognitive fatigue. This could improve exam scores in the short term but may lead to student frustration, disengagement, and high platform abandonment rates in the long run.